

Freight Rate Prediction

The objective is to predict the total freight price (posted_rate). The labeled data contains 48,000 loads from January to October 2025. Predictions are required for 12,000 loads from November and December.

Data-quality checks

Check	Training	Prediction data
Missing weight	300	165
Weight at or below zero	292	145
Missing market_index	374	249
Duplicate rows	0	0

Zero and negative weights are treated as missing. Numerical values are filled using training medians. Cleaning, scaling, and category encoding are fitted only on the training portion. Expensive loads are retained because their validity is not established.

Time-based validation

Stage	Training	Evaluation / prediction
Model selection	Jan-Jun: 28,806	Jul-Aug: 9,671
Final internal test	Jan-Aug: 38,477	Sep-Oct: 9,523
Final training	Jan-Oct: 48,000	Nov-Dec: 12,000

The split follows the dates because the assessment requires predictions for later months. July-August results determine the model choice. September-October is kept for the final internal evaluation.

Features describe the route, equipment, weight, and calendar. Full models also use market_index and quote_signal. Their origins are undocumented. Models estimate dollars per mile, then multiply by distance to obtain the total price.

Model Comparison and Test Results

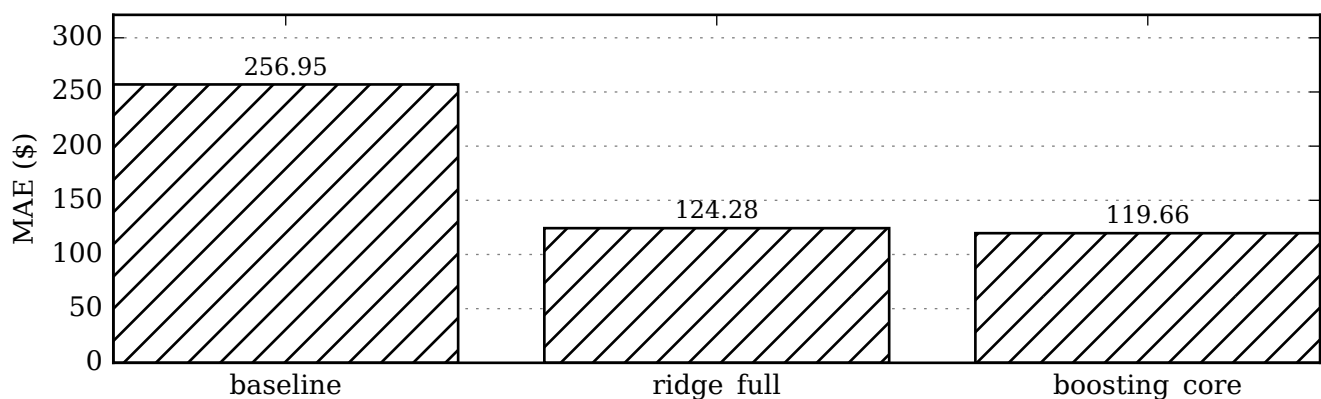
A distance baseline, Ridge regression, and histogram gradient boosting were compared. MAE measures average absolute dollar error. Lower values are better.

Model	Jul-Aug MAE	Jul-Aug RMSE
ridge_full	\$148.09	\$625.60
boosting_full	\$148.39	\$630.64
boosting_core	\$152.99	\$626.88
ridge_core	\$188.03	\$634.13
baseline	\$249.01	\$674.72

The main model is ridge_full, selected using July-August MAE. The December model is boosting_core, selected among models that do not require the two additional signals. The choices remain unchanged after the final test.

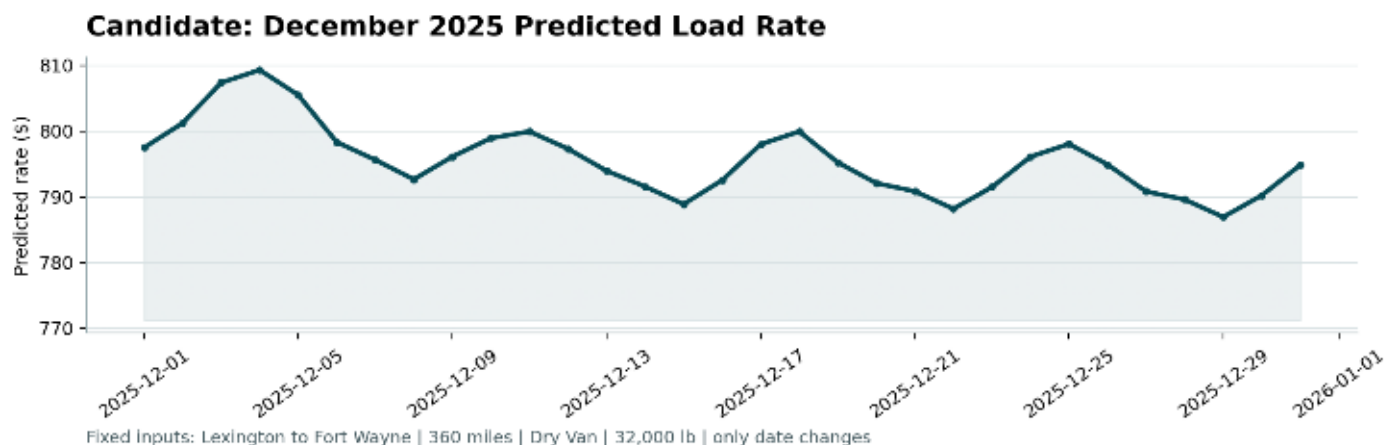
September-October internal test

Model	MAE	RMSE	MAPE
baseline	\$256.95	\$684.25	11.63%
ridge_full	\$124.28	\$638.13	5.30%
boosting_core	\$119.66	\$637.45	5.16%



December Predictions and Limitations

The fixed load is Lexington to Fort Wayne, 360 miles, Dry Van, and 32,000 lb. Only the date changes. Coordinates come from the labeled data, and the core model does not require `market_index` or `quote_signal`.



Predictions range from \$786.92 to \$809.29, averaging \$795.59. The calendar features produce the date-dependent changes. No November or December examples are available for training, so seasonality is uncertain.

Limitations

Large price outliers still cause substantial errors. Unfamiliar cities and changes in future market conditions may also affect performance. Further work could test more time windows, investigate large errors, and confirm how the supplied signals are produced.

The scorer checks file format and generates the chart. Final prediction accuracy is calculated by Spotter. The reported metrics are from the internal September-October test.

Sources: supplied assessment PDF, training and validation CSVs, prediction template, December inputs, and `score.py`.